We have seen that a population P which grows at a rate proportional to the size of the population satisfies the differential equation

$$\frac{dP}{dt} = kP. \tag{1}$$

We have seen previously that the general solution to Equation (1) is given by $P(t) = P(0)e^{kt}$, where $P(0)$ is an arbitrary real number (physically, we are only interested in the case $P(0) > 0$). Another way of writing Equation (1) is

$$\frac{1}{P}\frac{dP}{dt} = k \tag{2}$$

which says that the *relative growth rate* (the growth rate divided by the population size) is constant. Then Equation (2) says that a population with constant relative growth rate must grow exponentially.

As we discussed, a population often increases exponentially in its early stages but levels off eventually and approaches its carrying capacity because of limited resources. If $P(t)$ is the size of the population at time t, we assume that

$$\frac{dP}{dt} \approx kP \qquad \text{if } P \text{ is small}$$

This says that the growth rate is initially close to being proportional to size. But we also want to reflect the fact that the relative growth rate decreases as the population P increases and becomes negative if P ever exceeds its *carrying capacity* M, the maximum population that the environment is capable of sustaining in the long run. The simplest expression for the relative growth rate that incorporates these assumptions is

$$\frac{1}{P}\frac{dP}{dt} = k\left(1 - \frac{P}{M}\right).$$

Multiplying by P, we obtain the model for population growth known as the *logistic differential equation*:

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right). \tag{3}$$

We verified previously the following information directly from Equation (3):

- If P is small compared with M, then $\frac{P}{M} \approx 0$ and so $\frac{dP}{dt} \approx kP$.
- If $P \to M$, then $\frac{P}{M} \to 1$, so $\frac{dP}{dt} \to 0$.
- If $0 < P < M$, then $\frac{dP}{dt} > 0$ so the population increases.
- If $P > M$, then $1 - \frac{P}{M}$ is negative, hence $\frac{dP}{dt} < 0$ so the population decreases.

We now begin a more detailed analysis of the logistic differential equation.

1.) Equation (3) is separable. Find the general solution to the logistic differential equation

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right).$$

2.) Write the solution of the initial-value problem $\frac{dP}{dt} = 0.08P\left(1 - \frac{P}{1000}\right)$, $P(0) = 100$ and use it to find the population sizes $P(40)$ and $P(80)$. At what time does the population reach 900?

3.) In the 1930s the biologist G.F. Gause conducted an experiment with the protozoan *Paramecium* and used a logistic equation to model his data. The table below gives his daily count of the population of protozoa. He estimated the initial relative growth rate to be 0.7944 and the carrying capacity to be 64.

t (days)	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
P (observed)	2	3	22	16	39	52	54	47	50	76	69	51	57	70	53	59	57

Find the exponential and logistic models for Gause's data. Plot these in Desmos here: `https://www.desmos.com/calculator/rjyspi6gml` and comment on the fit.

4.) Biologists stocked a lake with 400 fish and estimated the carrying capacity (the maximal population for the fish of that species in that lake) to be 10,000. The number of fish tripled in the first year.

(a) Assuming that the size of the fish population satisfies the logistic equation, find an expression for the size of the population after t years.

(b) How long will it take for the population to increase to 5000?

5.) (a) Show that if P satisfies the logistic equation, then

$$\frac{d^2P}{dt^2} = k^2P\left(1-\frac{P}{M}\right)\left(1-\frac{2P}{M}\right).$$

(b) Deduce that a population grows fastest when it reaches half its carrying capacity.